University of North Georgia

Computer Science & Cybersecurity

CSCI 3250 Computer Security

General policies: Work in teams of two students. Each student submits the same file. Working as a solo cuts 10 points.

Project # 1.

Implementing and Understanding Scanning Using Python.

Description: In this project, students will develop Python code to connect to a server and scan which TCP ports are open. This project provides hands-on experience with ethical hacking practices. The project also emphasizes cybersecurity ethics, emphasizing the importance of controlled environments and legal considerations in penetration testing.

Specific Learning Outcomes:

1. By analyzing a complex computing problem and applying foundational cybersecurity principles, we were able to implement a practical solution that reflects the competencies outlined in ABET Outcome 1. In developing and debugging our code, we exercised software development fundamentals and applied core computer science theory.
2. (ABET Outcome 6). Additionally, the nature of the project—focused on security assessment and defense—allowed us to consider the ethical implications of port scanning practices and recognize professional responsibilities in computing.
3. (ABET Outcome 4). Through this project, we not only enhanced our technical knowledge but also demonstrated the ability to develop secure, responsible, and effective cybersecurity tools.

You are allowed to surf only for python syntax and for the questions at the end of this project but searching – including AI & ChatGPT- for the python solution is plagiarism. Copying code from the internet is plagiarism.

Python Program:

You are requested to write down a python script to scan your local machine 127.0.0.1. The functionality of your program is like the nmap functionality, but your script is not allowed to import the Python nmap library.

Your python script will be executed, from your local Machine, with the command:

```
python3  myNmap.py  127.0.0.1
```

The script must output the TCP open ports on the target (screenshot 1).

The script must catch any error like no inputting the ip address of the target (screenshot 2), inputting an ip address not reachable (screenshot 3)

Advise on how to develop this project:

1. Code in your own laptop.
2. For testing use your own ip address: python myNmap.py 127.0.0.1
3. Debug your program in your own environment.
4. Run your program as specified previously and take the corresponding screenshots.

Do a summary of the experience gained in this project, the problems encountered, how did you solve the problems and answer the following questions:

Questions:

1. How a client knows that a server has a tcp port open?
2. How a client knows that a server has a tcp port closed?
3. Did your script complete the three-way handshake? Why?
4. How a client knows that a server has a UDP port open?

What to submit:

1. A **pdf/Word file** that contains the following
    a. A cover sheet with
        i. Project number and name,
        ii. Course number, name, year, and semester,
        iii. Author's names and email addresses
        iv. Day and time
    b. A summary essay about the project in your own words, covering the following:
        i. The nature of the problem to be solved,
        ii. A description of how you solved this problem.
        iii. The issues and problems that you encountered and how did you solve those problems,
        iv. The concepts that you learned,
        v. The distribution of responsibilities and if those were completed.
        vi. References as appropriate. If there are no references, then you indicate explicitly the use of no references.
        vii. Screenshots as requested
        viii. The python code.

Grading Criteria:

Please read the Section What to Submit.

1. Submitting the project as a pdf/Word file:
    a. Cover sheet is 5 points.
    b. Summary essay 20 points including questions. Each question correctly answered is 5 points.
    c. Evidence the program runs as requested with no syntax

errors. 35 points. Each syntax error cost 10 points.

d. Screenshots as requested. 35 points.
e. Python code. 30 points.
f. Python code well documented/commented. 5 points.
g. Each student must submit the same pdf file in D2L. If pdf files submitted are different, the first one in the list will be the one considered. If a team member does not submit the pdf file, 30 points will be cut to that student.
h. Working independently will cut 10 points.
i. More than one file submitted will cut 10 points.
j. Different grade will be applied if a student must complete the whole project. Full credit for the student that complete correctly the project and zero for a student that did not fulfill responsibilities/deadlines.

2. There are two slip days for submitting the project.